International Review of Financial Analysis 109 (2026) 104803

Contents lists available at ScienceDirect

International Review of Financial Analysis

journal homepage: www.elsevier.com/locate/irfa

Supply chain finance digitalization and corporate risk-taking: Evidence from China

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ARTICLE INFO

Keywords:
Supply chain finance digitalization
Corporate risk-taking
Finance constraints
Supply chain concentration

ABSTRACT

Supply chain finance (SCF) digitalization that integrates emerging technologies into traditional SCF has attracted global attention amid widespread restructuring of industrial value chains. These shifts have increased the relevance of supply chain security and corporate risk-taking. Although prior research has examined SCF digitalization, its impact on corporate risk-taking remains underexplored. Using data on Chinese listed firms from 2011 to 2023, SCF digitalization promotes corporate risk-taking by easing financial constraints and reducing supply chain concentration. The effect is significant among small and medium-sized enterprises, growth-stage firms, and nonstate-owned enterprises and is further amplified under China's supply chain innovation pilot policy.

1. Introduction

Technological advancements in high-tech industries are intensifying global competition across geopolitics, technology, and trade, reaching unprecedented levels of complexity. Similarly, industrial and supply chains are undergoing broad restructuring, which is increasingly crystallizing into distinct yet interwoven trajectories and reshaping development paths and competitive dynamics worldwide. This restructuring amplifies credit, operational, and liquidity risks along supply chains, in which volatility at a core enterprise can trigger a "domino effect" of financing defaults among small and medium-sized enterprises (SMEs; Geng et al., 2024), posing significant challenges to traditional risk-control mechanisms. Meanwhile, rapid digital innovation is transforming the configuration and management of supply chains and their financial activities, bringing digitalization of supply chain finance (SCF) and shifts in corporate risk-taking to the forefront for academia and industry.

Regarding SCF digitalization, Song et al. (2020) showed that integrating physical and financial flows creates a tightly connected ecosystem with enhanced operational linkages and low-friction data exchange. Traditional SCF faces persistent weaknesses such as information asymmetry and systemic vulnerability, in which a single-node default can trigger chain-wide contagion. Digital technologies stabilize supply chain relationships (Zhou et al., 2024) and improve the risk-taking and operational capabilities of node enterprises (Pan, Pan, et al., 2020; Zhang, Hu, et al., 2021). By embedding digital technologies into SCF practices (Gong et al., 2022), enterprises gain real-time access to information across the chain. Risk alerts at one node allow others to adjust investment and financing decisions promptly, thereby limiting contagion. Prior studies examined SCF digitalization's effects on innovation (Lei et al., 2024) and reduced financial fraud (Jiang et al., 2024), yet research on its influence over corporate risk-taking remains limited.

Corporate risk-taking reflects firms' pursuit of higher profits by engaging in risk-taking activities (Boubakri et al., 2013). It is vital for value creation (John et al., 2008), capital structure decisions (Chen et al., 2017), and resource allocation (Faccio et al., 2016), helping firms seize investment opportunities, enhance performance, and maintain competitive edge (Dao & Phan, 2023; Koirala et al., 2020). At the macro level, it supports technological innovation and efficient resource allocation, contributing to sustainable economic growth (Su et al., 2017). Amid uncertainty, firms can rationally assume reasonable risks to pursue higher expected returns (Huang et al., 2021). However, limited evidence exists on whether SCF digitalization promotes corporate risk-taking or the mechanisms through which it may influence firms' risk behavior.

Corporate risk-taking is influenced by internal characteristics and external conditions. Internally, key determinants include managerial traits (Kim & Lu, 2011), corporate governance (Chu et al., 2020; Huang et al., 2021; John et al., 2008; Lan et al., 2025), entrepreneur demographics (Beber & Fabbri, 2012; Faccio et al., 2016; Huang et al., 2021; Malmendier & Nagel, 2011), managerial incentives (Low, 2009),

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https://doi.org/10.1016/j.irfa.2025.104803
Received 23 June 2025; Received in revised form 21 November 2025; Accepted 22 November 2025
Available online 22 November 2025
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and digital transformation (Wu & Wang, 2024). Externally, financial regulation (Liu & Wu, 2024), market conditions (Sheikh, 2019), fintech development (Ding et al., 2022), SCF (Liu et al., 2025), and environmental factors (Li et al., 2025; Tran & Le, 2022; Zhang, Yang, & Liu, 2021) also play critical roles. Research in the digital economy highlights the interaction between fintech and supply chain ecosystems: leveraging text analysis of 3.2 million announcements, Xu et al. (2025) find that fintech-enabled SCF significantly increases corporate risk-taking. However, micro-level evidence on how SCF digitalization affects corporate risk behavior remains scarce. This study addresses that gap by examining the relationship between SCF digitalization and corporate risk-taking, offering the first empirical analysis on this topic.

In China, rapid deployment of artificial intelligence, along with advances in large language model auditing since 2023, has heightened corporate awareness of the need for reliable, automated digital systems. This shift has fueled the rapid growth of the SCF technology market, where total transaction value increased from RMB 5.2 T in 2020 to RMB 11.7 T in 2024, with a compound annual growth rate of 22.2 %. Against this backdrop, this article analyzes the unique characteristics of China's listed firms to uncover an underexamined dimension of SCF finance.

This article contributes to the literature in several ways. First, it broadens understanding of the economic effects of SCF digitalization beyond previously documented outcomes—such as green innovation (An et al., 2024), reduced financial fraud (Jiang et al., 2024), and regional innovation (Zheng & Xiao, 2025)—by examining its influence on corporate risk-taking, an area with scarce empirical evidence. Specifically, this study demonstrates that SCF digitalization reduces financial constraints and lowers supplier concentration, thereby promoting corporate risk-taking. By clarifying these mechanisms, the findings deepen knowledge of how SCF digitalization enables risk-taking within China's industrial chain.

Second, this study enriches the literature on the determinants of corporate risk-taking. Prior research has largely focused on internal characteristics such as board fault lines (Pu et al., 2023), capital structure mismatch (Li & Zhou, 2025), enterprise financialization (Zhao et al., 2025), chief executive officer (CEO) traits (Huang et al., 2021; Benmelech & Frydman, 2015), and enterprise, social, and governance performance (He et al., 2015), along with external determinants including financial regulations (Liu & Wu, 2024), fintech development (Tang et al., 2024), industrial robots (Pu et al., 2024), carbon policies (Xing et al., 2023), climate risk (Li et al., 2025), economic policy uncertainty (Zhang, Yang, & Liu, 2021), and green finance (Deng et al., 2023). Given the limited evidence on SCF digitalization, we establish its relevance to corporate risk-taking within supply chain networks, highlighting how financing embedded in digitally enhanced structures shapes firms' risk behavior.

Finally, this research complements and extends the empirical literature by proposing a novel approach for assessing the development level of SCF digitalization. Existing indicators primarily assess SCF digitalization from the financial service provision perspective (Tong & Wang, 2024; Liu et al., 2023) and are constructed at the provincial level. In contrast, this study develops an enterprise-level indicator system, broadening the analytical scope of SCF digitalization as a novel development paradigm.

This study addresses a previously overlooked area by examining the understudied relationship between SCF digitalization and corporate risk-taking. Positioned at the intersection of SCF digitalization, corporate governance, and supply chain management, the study findings advance scholarly understanding and offer practical implications for researchers and practitioners.

2. Hypothesis development

Corporate risk-taking reflects an enterprise's strategic inclination to pursue high-uncertainty investments for value maximization (Tran, 2019). SCF alleviates financing and liquidity constraints by providing tailored solutions to core firms and their upstream suppliers (Rahman et al., 2024), enhancing suppliers' capacity to fulfill obligations and reducing disruption risk. Fintech enabled SCF further strengthens firms' risk-taking capabilities by expanding investment scope and improving capital allocation efficiency (Xu et al., 2025), thereby supporting supply chain resilience amid volatility. Risk-tolerant firms collaborating with financially constrained suppliers are more inclined to adopt SCF, gaining real-time monitoring through digital risk-control models (Wang, 2025). This reduces the cost of risk-taking (Zhao & Lei, 2024), forming a virtuous cycle in which SCF digitalization lowers risk costs and increases willingness to take risks (Han & Wang, 2025; Pan, Xu, et al., 2020). As a result, SCF enhances access to financial resources, optimizes capital allocation, and improves supply chain efficiency, positioning corporate risk-taking as a key driver of supply chain security.

Traditional SCF relies on operational data streams—logistics, capital flows, and information flows—generated from transactions between core firms and their upstream and downstream partners. These data streams reflect production and operational status and form the basis of SCF service offerings (Song & Chen, 2016). New digital technologies such as artificial intelligence, blockchain, and cloud computing expand financial services' operational and strategic boundaries (Dewan & Kraemer, 2000; Tang et al., 2020; Xu et al., 2025). SCF digitalization represents an innovative network-based integration of the industrial and financial processes through networked digital tools (Caniato et al., 2016), enabling full transparency and traceability across procurement, production, logistics, and sales. Improved transparency reduces information asymmetry and the risks associated with incomplete or inaccurate data (Wu, Xu, & Lin, 2021). Moreover, digitalization converts soft information into quantifiable "hard information" (Cenni et al., 2015), allowing more sophisticated assessment of enterprise operations and market behavior (Einav & Levin, 2014). Therefore, we proposed Hypothesis 1.

H1. SCF digitalization advances corporate risk-taking.

A fundamental prerequisite for effective collaboration between supply chain entities is a foundation of trust (Bouncken et al., 2020). Trust supports the provision of trade credit financing (Jiang et al., 2022) and increases financial institutions' credit willingness. SCF digitalization enhances the availability and distribution of financial services (Duncombe & Boateng, 2009) and improves operational transparency by incorporating unstructured information into market evaluation. This creates a rich database of "soft information" (Huang et al., 2021) and facilitates credit risk assessment (Laeven et al., 2015), fostering a business environment conducive to growth. By lowering financing thresholds and integrating SMEs more deeply into supply chains, SCF digitalization expands access to diverse financing channels. Increased funding availability helps overcome capital constraints in high-risk activities such as R&D (Hu et al., 2025) and market expansion, accelerating fund circulation, improving allocation efficiency, and enhancing firms' resilience to market volatility.

SCF digitalization generates substantial digital transaction data and transforms traditionally intangible "soft information" from production and operational activities into analyzable content (Mocetti et al., 2017). The evolving features of supply chains—networked sharing, intelligent collaboration, and big-data usage (Liu et al., 2023)—are reshaping cooperation among upstream and downstream partners. Digital technologies enable cross-organizational connectivity, ensuring smoother flows of production factors (Conceição et al., 2001). Integrated data elements broaden the information used in supply chain credit assessments, allowing more accurate identification of risk exposure throughout transactions (Song et al., 2022). These advances stabilize supply chain relationships and strengthen firms' risk-management capabilities (Kwak et al., 2018). They also enhance collaborative efficiency, reduce upstream and downstream concentration, and diversify supply chain structures, helping enterprises disperse operational risk (Wu & Yao, 2023). Importantly, lower concentration mitigates

Table 1
Index system for supply chain finance (SCF) digitalization.

Dimension	Indicator	Indicator description	References	Properties
SCF	SCF level: A1	Extracting keywords, such as SCF from the text of annual reports of listed companies based on word frequency	Zhou and Wu (2022)	+
	Financial oriented SCF: A2	Accounts receivable/operating income	Li et al. (2019), Liu, Wang, and Shou (2020)	+
	Supply chain oriented SCF: A3	Accounts receivable holding period + Inventory holding period − Accounts payable payment period	Cosci et al. (2020)	+
Digitalization	Cognition of digital finance: A4	Extracting keywords, such as digital finance from the text of annual reports of listed companies based on word frequency	Wang and Xie (2021)	+
	Enterprise digital transformation: A5	Extract keywords, such as artificial intelligence technology, big data technology, cloud computing technology, blockchain technology, and digital technology application from the text of annual reports of listed companies.	Wu, Hu, et al. (2021)	+
	Digital investment: A6	The proportion of investments in intangible assets and fixed assets related to digital technology in the detailed breakdown of fixed assets and intangible assets, as a percentage of the enterprise's total annual assets.	Qi et al. (2020)	+

"relationship-specific investment" risks (Lu et al., 2025), decreasing over-reliance on key partners and enabling more flexible strategic decision-making. Therefore, we propose Hypothesis 2.

H2. SCF digitalization increases corporate risk-taking by easing financial constraints and reducing supply chain concentration.

3. Research design

3.1. Data

This study examines Chinese A-share listed companies during 2011–2023. To ensure data quality and mitigate potential bias, we implemented the following screening rules:

- Removed ST/*ST and delisted firms;
- Excluded financial and insurance institutions;
- Deleted observations with missing or abnormal financial data; and
- Winsorized continuous variables at the 1 % level.

The final sample included 21,283 firm-year observations, with financial data collected from the CSMAR database.

3.2. Model

To examine the impact of SCF digitalization on corporate risk-taking behavior, we estimated the following model:

$$Risk_{i,t} = \alpha_0 + \alpha_1 SCFD_{i,t} + \delta Controls_{i,t} + + u_i + \theta_t + \varepsilon_{i,t} \quad (1)$$

where $Risk_{i,t}$ denotes corporate risk-taking, $SCFD_{i,t}$ measures the level of supply chain finance digitalization (SCFD), $Control_{i,t}$ represents control variables, u_i and θ_t indicate firm and year fixed effects, and $\varepsilon_{i,t}$ is the error term.

Next, we estimate Model (2) to test whether SCFD affects corporate risk-taking (*Risk*) through reducing financial constraints:

$$Risk_{i,t} = \gamma_0 + \gamma_1 SCFD_{i,t} + \delta Controls_{i,t} + + u_i + \theta_t + . \quad (2)$$

We then estimated Model (3) to evaluate the mediating role of supply chain concentration (*SCC*) in the relationship between SCFD and corporate risk-taking:

$$Risk_{i,t} = \gamma_0 + \gamma_1 SCFD_{i,t} + \delta Controls_{i,t} + + u_i + \theta_t + \varepsilon_{i,t} \quad (3)$$

3.3. Variables

3.3.1. SCFD

SCFD is an advanced, digitally enhanced form of traditional SCF, characterized by blockchain-based recording of enterprise transaction data. It leverages data elements to improve the efficiency and flow of key production factors (Zhang, 2022) and enables end-to-end traceability of material, information, and capital flows (Gong et al., 2022). Supported by strong big-data processing and intelligent infrastructure, SCF digitalization enhances service accessibility and value, enabling flexible and efficient interorganizational integration (Büyüközkan & Göçer, 2018). This stage reflects the deep integration of enterprise digitalization with SCF. We use the entropy method to measure the level of SCF digitalization across two dimensions—SCF and digitalization—summarized in Table 1.

3.3.1.1. SCF dimension. SCF practices are categorized into finance- and supply chain-oriented SCF (Gelsomino et al., 2016). Finance-oriented SCF focuses on monitoring transaction-level information for individual firms or specific supply chain segments, such as credit status and asset conditions (Song et al., 2021). In contrast, supply chain-oriented SCF emphasizes comprehensive oversight of the entire operational process across pre- and post-transaction stages (Song & Lu, 2017); this approach integrates upstream and downstream data and prioritizes risk mitigation through supply chain credit evaluation (Moretto et al., 2019). This dimension is assessed through three perspectives:

- SCF level (A1). Following Zhou and Wu (2022), we measured supply chain finance level by extracting the frequency of SCF-related keywords in firms' annual reports.
- Finance-oriented SCF level (A2). Based on Liu, Wang, and Shou (2020) and Li et al. (2019), this indicator uses the proportion of revenue tied up in receivables, reflecting financing support from core firms to downstream partners (Song et al., 2021):
- Supply chain-oriented SCF level (A3). Consistent with Cosci et al. (2020), this measure is calculated as follows: A3 = accounts receivable holding period + inventory holding period − accounts payable payment period.

3.3.1.2. Digitalization dimension. This dimension includes three indicators: corporate digital finance awareness (A4), corporate digitalization (A5), and corporate digital investment (A6).

- Corporate digital finance awareness (A4). Following Wang and Xie (2021), we measured the frequency of digital finance–related keywords such as "internet" and "digitalization" per 10,000 words in firms' annual reports.
- Corporate digitalization (A5). Based on Wu, Hu, et al. (2021), this indicator captures keyword frequency related to big data, AI, cloud computing, blockchain, and other digital applications in annual reports.

- Digital investment (A6). Consistent with Zhang and Li (2023) and Qi et al. (2020), this measure reflects the share of digital technology–related intangible and fixed asset investment in firms' total assets:

3.3.2. Corporate RISK-taking (RISK)

Following Tang et al. (2024) and John et al. (2008), this study constructed two measures of corporate risk-taking. *Risk_1* is the standard deviation of the rolling 3-year industry-adjusted ROA, and *Risk_2* is the range of its rolling 3-year industry-adjusted period:

$$AdjROA_{i,t} = ROA_{i,t} - \frac{1}{n}\sum\nolimits_{t-1}^{n} ROA_{i,t} \tag{4}$$

$$SD_ROA_{i,t} = \sqrt{\frac{1}{T-1}\sum_{t=1}^{T}\left(Adj_ROA_{i,t} - \frac{1}{T}\sum_{t=1}^{T} Adj_ROA_{i,t}\right)^2}, T = 3 \tag{5}$$

$$ROA_Diff_{i,t} = Max\left(Adj_ROA_{i,t}\right) - Min\left(Adj_ROA_{i,t}\right) \tag{6}$$

ROA represents the return on total assets, where subscripts *i* and *t* indicate the supplier firm and year, respectively. *Adj_ROA* denotes the industry-adjusted return on assets, calculated as the firm's ROA minus the annual industry average. *SD_ROA* is the standard deviation of *Adj_ROA* over rolling 3-year windows (e.g., 2008–2010, 2009–2011), and *ROA_Diff* is the difference between the maximum and minimum *Adj_ROA* within the same intervals. We also constructed two customer concentration metrics based on Patatoukas (2012) and Campello and Gao (2017). The first measure is the share of total sales attributed to the top five customers (*CustomerSales*).

3.3.3. Control variables

Drawing on established research on corporate risk-taking, we include several control variables capturing operational characteristics: firm size (*Size*), firm age (*Age*), the ratio of net fixed assets to total assets (*Fixed*), operating income growth rate (*Growth*), a loss indicator (*Loss* = 1 if net profit <0, else 0), CEO duality (*Dual*), operating cash flow ratio (*Cashflow*), market-to-book ratio (*MB*), and ownership concentration (*TOP10*).

4. Empirical results

4.1. Descriptive statistics

Table 2 presents the descriptive statistics of all variables. The average level of *Risk1* was 0.0253, with a median of 0.0160, a maximum value of 0.2207, and a minimum value of 0.0009. The mean value of *Risk2* was 0.0463, with a standard deviation of 0.0546, a maximum value of 0.3687, and a minimum value of 0.0000. These findings show substantial divergence in risk levels, indicating variation in corporate risk-taking across firms. The average SCFD value was 0.1183, with a standard deviation of 0.1105, indicating certain gap in SCFD development levels among firms. The descriptive statistical results for the control variables also show clear heterogeneity, supporting the need to include them in the regression models for robust empirical testing.

4.2. Main results

Initially, we estimated model (1) to test whether SCFD promotes corporate risk-taking (**H1**). Year and firm fixed effects control for temporal trends and unobservable firm characteristics. Table 3 reports the stepwise regression results. From columns (1) and (2) of Table 3, when only the dependent variable, *Risk*, and the key explanatory variable, *SCFD*, were included, the coefficients of SCFD are positive and significant at the 1 % level. After adding control variables, columns (3) and (4) show that the coefficients of SCFD remain positive and significant at the 1 % level. Therefore, SCFD significantly increases firms' risk-taking levels, empirically supporting **H1**.

4.3. Robustness tests

Although the baseline results support **H1**—that SCFD encourages corporate risk-taking—potential endogeneity concerns persist, including omitted variable bias, reverse causality, and sample selection issues. These factors could affect the reliability of the findings. To mitigate such risks and strengthen the robustness of the conclusions, we adopted multiple econometric strategies designed to mitigate endogeneity and enhance the validity and reliability of the baseline regression conclusions.

4.3.1. IV methods

SCFD enhances the risk-taking capacity of supply chain enterprises. Meanwhile, increased risk-taking can further promote the expansion of the SCFD ecosystem, suggesting potential reverse causality. Therefore, we conducted an instrumental variable approach. Specifically, we use the mean SCFD of firms in the same city and industry—*SCFD_IND* and *SCFD_REG*—as instrumental variables.

Table 4 presents the IV and 2SLS estimation results. The LM test yields $p = 0$, and the first-stage F-values exceed critical thresholds; additionally, the Sargan test is passed, confirming instrument validity. Columns (1) and (2) denote that *SCFD_IND* and *SCFD_REG* are positive and significant at the 1 % level (0.7534 and 0.7812, respectively). Columns (3) and (4) demonstrate that the coefficients of *SCFD_IND* on *RISK1* and *RISK2* are 0.0608 and 0.1017, respectively, which is positive and significant at the 1 % level, and columns (5) and (6) show that the coefficients of *SCFD_REG* on *RISK1* and *RISK2* are 0.0277 and 0.0489, respectively, suggesting similarly significant positive effects. Therefore, after controlling for endogeneity issues using instrumental variables, the positive relationship between SCFD and corporate risk-taking remains robust.

4.3.2. Propensity score matching method (PSM)

To enhance the reliability of the regression results, we performed a PSM robustness test. Based on the annual industry median of SCFD, firms were divided into high- and low-SCFD groups. Using the baseline regression control variables as covariates, 1:1 nearest-neighbor matching was conducted, and the matched samples were re-estimated. After matching, both samples met the equilibrium hypothesis, with no significant differences in covariates between groups. From Table 5, the coefficients of *SCFD* remained significant at the 1 % and 5 % levels, confirming the robustness of our main conclusions.

4.3.3. Lagging one-period processing

To further mitigate potential endogeneity from bidirectional causality, we lagged all explanatory variables by one period. Columns (1) and (2) of Table 6 show that the coefficients of *SCFD* remained

Table 2
Summary statistics.

	N	Mean	Std. Dev.	Min	Median	Max
SCFD	21,283	0.1183	0.1105	0.0003	0.0879	0.7326
Risk1	21,283	0.0253	0.0307	0.0009	0.0160	0.2207
Risk2	21,283	0.0463	0.0546	0.0000	0.0295	0.3687
TOP10	21,283	0.5793	0.1744	0.0000	0.5981	0.9081
Fixed	21,283	0.2133	0.1448	0.0034	0.1868	0.6486
Dual	21,283	0.2984	0.4576	0.0000	0.0000	1.0000
Size	21,283	22.4243	1.2928	20.5035	22.2156	26.4973
Growth	21,283	0.1598	0.3146	−0.4656	0.1125	1.6878
MB	21,283	0.6304	0.2465	0.1379	0.6283	1.1914
Cashflow	21,283	0.0547	0.0651	−0.1244	0.0522	0.2440
Age	21,283	2.0512	0.7892	0.0000	2.0794	3.3322
Loss	21,283	0.0818	0.2741	0.0000	0.0000	1.0000

Note: Std. Dev. means Standard Deviation.

Table 3
Baseline regression results.

	(1)	(2)	(3)	(4)
Variable	*Risk1*	*Risk2*	*Risk1*	*Risk2*
SCFD	0.0159***	0.0296***	0.0146***	0.0258***
	(0.0044)	(0.0078)	(0.0042)	(0.0074)
TOP10			−0.0081***	−0.0126***
			(0.0018)	(0.0032)
fixed			−0.0131***	−0.0200***
			(0.0036)	(0.0064)
Dual			−0.0002	−0.0001
			(0.0008)	(0.0013)
Size			−0.0037***	−0.0063***
			(0.0008)	(0.0013)
Growth			−0.0023***	−0.0036**
			(0.0009)	(0.0015)
MB			−0.0084***	−0.0137***
			(0.0015)	(0.0027)
Cashflow			0.0149***	0.0248***
			(0.0043)	(0.0076)
Age			0.0063***	0.0189***
			(0.0009)	(0.0016)
Loss			0.0280***	0.0507***
			(0.0011)	(0.0020)
Constant	0.0235***	0.0428***	0.1028***	0.1605***
	(0.0005)	(0.0010)	(0.0165)	(0.0290)
Observations	21,283	21,283	21,283	21,283
R-squared	0.4075	0.4036	0.4637	0.4658
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Adjusted R^2	0.314	0.309	0.379	0.381

Note: ***, **, and * demonstrate significance at the 1 %, 5 %, and 10 % levels, respectively.

significantly positive at the 1 % level, indicating that the results are still robust after considering temporal delays in *SCFD*'s effect on corporate risk-taking.

4.3.4. Excluding the impact of major exogenous shocks

Major exogenous shocks can strongly influence firms' investment and financing decisions. To isolate the incremental effect of *SCFD* on corporate risk-taking from broader macroeconomic and other factors, we examined the COVID-19 pandemic of 2020 and 2021 as a potential exogenous shock that could influence the regression outcomes within the sample period. To mitigate the potential effects of these external shocks, we performed a benchmark estimation after excluding observations from these years. Columns (3) and (4) of Table 6 show that the coefficient of *SCFD* remains significantly positive at the 1 % level, demonstrating that *SCFD* contributes to enhancing corporate risk-taking even when pandemic years are excluded. Therefore, our findings are resilient to major exogenous disturbances.

4.3.5. Replacement of explained variable

To further ensure robustness, we replicated the main regressions using alternative measures of corporate risk-taking. First, we extended the rolling window to 5 years and defined *Risk3* as the standard deviation of industry-adjusted ROA over a rolling 5-year period. Additionally, following Tang et al. (2024), we constructed *Risk4* and *Risk5* as alternative measures of risk-taking by calculating *ROAEBIT*, defined as *EBIT* divided by total assets (Boubakri et al., 2013), and computed its standard deviation and range over rolling windows. Further, consistent with Kini and Williams (2012) and Huang et al. (2021), we defined *Risk6* using the ratio of R&D expenditure to operating revenue, reflecting the high-risk nature of innovation activity. Therefore, R&D investment expenditure reflects the risk-taking level of enterprises to a certain extent. From Table 7, across all alternative measures, the coefficients of *SCFD* remain positive and significant at the 1 % level, consistent with our baseline results and confirming robustness.

4.3.6. Weaken the influence of missing variables

Although the baseline model includes multiple controls, omitted variables may still affect the relationship between SCFD and corporate risk-taking (Dou et al., 2024). Therefore, we added provincial fixed effects using the registered location of each enterprise and include

Table 5
Propensity scores matching method.

Variable	(1) *Risk1*	(2) *Risk2*
SCFD	0.0123**	0.0214**
	(0.0059)	(0.0104)
Constant	0.1236***	0.1953***
	(0.0225)	(0.0399)
Controls	Yes	Yes
Year FE	Yes	Yes
Firm FE	Yes	Yes
Observations	11,202	11,202
Adjusted R^2	0.371	0.373

Note: ***, **, and * demonstrate significance at the 1 %, 5 %, and 10 % levels, respectively.

Table 4
Instrumental variable approach.

Variable	(1)	(2)	(3)	(4)	(5)	(6)
	First stage		Second stage (IV = *SCFD_IND*)		Second stage (IV = *SCFD_REG*)	
	SCFD	*SCFD*	*Risk1*	*Risk2*	*Risk1*	*Risk2*
SCFD_IND	0.7534***					
	(0.0272)					
SCFD_REG		0.7812***				
		(0.0157)				
SCFD			0.0608***	0.1017***	0.0277***	0.0489***
			(0.0200)	(0.0350)	(0.0096)	(0.0172)
Cragg–Donald Wald F statistic	956.463	2592.768				
LM statistics	672.600	1010.400				
	(0.000)	(0.000)				
10 % maximal 16.38 IV size	16.38					
Constant	−0.3402***	−0.3128***				
	(0.0316)	(0.0310)				
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	21,283	21,283	21,283	21,283	21,283	21,283
Adjusted R^2	0.787	0.804	−0.0578	−0.0452	−0.0482	−0.0371

Note: ***, **, and * demonstrate significance at the 1 %, 5 %, and 10 % levels, respectively.

Table 6
Lagged SCFD variables and excluding sample.

Variable	(1)	(2)	(3)	(4)
	Lagged		Excluding Sample	
	Risk1	*Risk2*	*Risk1*	*Risk2*
L.SCFD	0.0145***	0.0264***		
	(0.0048)	(0.0084)		
SCFD			0.0103**	0.0184**
			(0.0046)	(0.0081)
Constant	0.1273***	0.2221***	0.0886***	0.1380***
	(0.0196)	(0.0346)	(0.0178)	(0.0314)
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	16,975	16,975	17,483	17,483
Adjusted R^2	0.407	0.412	0.337	0.342

Note: ***, **, and * demonstrate significance at the 1 %, 5 %, and 10 % levels, respectively.

industry fixed effects with industry dummy variables to control for sector-specific factors. Table 8 reports the results. After controlling for regional effects, the coefficients of *SCFD* are 0.0141 and 0.0248, both significant at the 1 % level, consistent with the baseline estimates. When controlling for industry effects, the coefficients become 0.0146 and 0.0258, which were significant at the 1 % level. Therefore, *SCFD* continues to positively influence corporate risk-taking even with regional and industry fixed effects, thereby reinforcing the robustness of our conclusions.

5. Further analysis

5.1. Heterogeneity analysis

This study evaluates heterogeneity in the effects of SCFD from three perspectives: firm size, age, and ownership structure. First, based on information asymmetry theory, digitalization alleviated information silos in supply chains, further alleviating financing constraints and enhancing risk-taking capacity (Wang et al., 2025). However, information transparency differs between state- and nonstate-owned enterprises, so the impact of SCFD varies by ownership type. Second, from a resource-based view, large firms possess strong digital infrastructure (e.g., ERP systems) and can integrate SCFD tools (e.g., blockchain supply chain platforms) efficiently (Lu et al., 2025), while SMEs often depend on third-party platforms because of limited resources, affecting their ability to bear independent risks. Thus, firm size may moderate SCFD's effect on risk-taking. Finally, according to signaling theory, young firms may lack collateral and rely more on SCFD-generated transaction data as credit signals to secure financing, whereas mature firms already possess strong reputations. Therefore, SCFD will have a strong impact on younger firms' risk-taking.

Panel A of Table 9 demonstrates that the coefficients of SCFD are larger and significant for small firms, but not significant for large firms. Panel B indicates that SCFD has a stronger and more significant effect in younger firms. Panel C further shows that coefficients for SCFD are significantly positive and larger in nonstate-owned enterprises (SOEs) compared with SOEs.

Therefore, smaller, younger, and nonstate-owned firms benefit more from SCF digitalization. SCFD significantly increases the risk-taking propensity of SMEs, growth-stage firms, and non-SOEs, reflecting its nature as an inclusive financial tool that supports firms with greater financing constraints and development needs.

Table 7
Alternative measure of the explanatory variable.

Variable	(1)	(2)	(3)	(4)
	Risk3	*Risk4*	*Risk5*	*Risk6*
SCFD	0.0111***	0.0138***	0.0324***	0.0400***
	(0.0030)	(0.0039)	(0.0078)	(0.0054)
Constant	0.0555***	0.1186***	0.2193***	0.1121***
	(0.0114)	(0.0145)	(0.0296)	(0.0110)
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	21,283	21,022	21,022	17,230
Adjusted R2	0.530	0.392	0.529	0.437

Note: ***, **, and * demonstrate significance at the 1 %, 5 %, and 10 % levels, respectively.

Table 8
Change the fixation effect.

	(1)	(2)	(3)	(4)
Variable	*Risk1*	*Risk2*	*Risk1*	*Risk2*
SCFD	0.0141***	0.0248***	0.0146***	0.0258***
	(0.0042)	(0.0074)	(0.0042)	(0.0074)
Constant	0.1030***	0.1611***	0.1028***	0.1605***
	(0.0164)	(0.0290)	(0.0165)	(0.0290)
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	No	No
Industry FE	No	No	Yes	Yes
Observations	21,279	21,279	21,283	21,283
Adjusted R^2	0.383	0.385	0.379	0.381

Note: ***, **, and * demonstrate significance at the 1 %, 5 %, and 10 % levels, respectively.

5.2. Mechanism tests

This study further examine the mechanisms through which SCFD influences corporate risk-taking. We propose that SCFD enhances firms' risk-taking by reducing financial constraints and lowering SCC. To test these channels, we use *FC* as a proxy for financial constraints and construct two dummy variables per Dang Viet & PhanHieu (2016), as *FC* more accurately reflects cash flow pressure. In addition, the proportions of purchases from the top five suppliers and sales to the top five customers for each firm (Tang, 2009) are used to construct a dummy variable *SCC*, which equals 1 if the firm's concentration exceeds the median and 0 otherwise.

Table 10 reports the mediation results. In columns (1)–(3), the coefficients of *SCFD* on financing constraints is −0.0344, on *Risk1* is 0.0140, and on *Risk2* is 0.0247, respectively, all significant at the 1 % level. This suggests that SCFD promotes corporate risk-taking by easing financial constraints. In columns (4)–(6), the coefficients of *SCFD* on SCC is −6.6543, on *Risk1* is 0.0149, and on *Risk2* 0.0262, respectively, again significant at the 1 % level. These findings indicate that SCFD also increases risk-taking by reducing SCC. Thus, both mechanisms hold, providing empirical support for **H2**.

5.3. Moderation effect tests

Since 2018, China has implemented pilot policies promoting supply chain innovation within selected cities and enterprises. These pilots provide a favorable environment for SCFD development by strengthening policy support, improving digital and logistical infrastructure, and cultivating industrial ecosystems. As a result, the pilot policy enhances the effectiveness of SCFD.

Pilot enterprises benefit from preferential measures, including subsidies, tax incentives, and special funds that lower the cost of building digital platforms (Shen & Qiao, 2024). Such initiatives encourage greater participation in the SCFD ecosystem, facilitating broader involvement in supply chain activities, improving financing access, and reducing excessive supplier or customer concentration.

Additionally, local governments in supply chain innovation pilot

Table 9
Heterogeneity analysis.

Panel A: Effect of different firm sizes				
Variable	Large firm		Small firm	
	(1)	(2)	(3)	(4)
	Risk1	*Risk2*	*Risk1*	*Risk2*
SCFD	−0.0031	−0.0045	0.0386***	0.0669***
	(0.0048)	(0.0085)	(0.0074)	(0.0130)
Constant	0.1335***	0.2284***	0.1945***	0.3222***
	(0.0233)	(0.0421)	(0.0419)	(0.0719)
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
p	0.0000			
Observations	10,611	10,611	10,388	10,388
Adjusted R^2	0.420	0.419	0.421	0.427
Panel B: Effect of different firm ages				
	Mature firm		Young firm	
Variable	(1)	(2)	(3)	(4)
	Risk1	*Risk2*	*Risk1*	*Risk2*
SCFD	0.0012	0.0026	0.0122**	0.0209**
	(0.0056)	(0.0099)	(0.0060)	(0.0104)
Constant	0.0725***	0.1178***	0.0869***	0.1195**
	(0.0226)	(0.0413)	(0.0310)	(0.0534)
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
p	0.0000			
Observations	10,212	10,212	10,810	10,810
Adjusted R^2	0.485	0.483	0.458	0.458
Panel C: Effect of different firm properties				
	State-owned firms		Nonstate-owned firms	
Variable	(1)	(2)	(3)	(4)
	Risk1	*Risk2*	*Risk1*	*Risk2*
SCFD	−0.0054	−0.0104	0.0241***	0.0428***
	(0.0066)	(0.0115)	(0.0054)	(0.0094)
Constant	0.0841***	0.1344***	0.1235***	0.1921***
	(0.0210)	(0.0378)	(0.0227)	(0.0399)
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
p	0.0000			
Observations	7015	7015	14,268	14,268
Adjusted R^2	0.347	0.346	0.386	0.390

Note: ***, **, and * demonstrate significance at the 1 %, 5 %, and 10 % levels, respectively.

areas often lead the development of unified service platforms to promote coordinated data sharing across departments. As Goldfarb and Tucker (2019) emphasized, data derive full value through connectivity and integration. Acting as a "connector" and an "integrator," these public platforms advance SCF digitalization, improve information transparency, lower information costs, strengthen risk assessment, and ultimately increase firms' risk-bearing capacity.

Geographic proximity and policy guidance enhance trust and collaboration, resulting in more mature and industry-specific SCFD solutions and reinforcing risk cogovernance. Accordingly, this section proposes the following hypothesis:

H3. Supply chain innovation policy positively moderates the effect of SCFD on corporate risk-taking.

The moderating variable "supply chain innovation" (*DID*) is constructed following Shen and Qiao (2024) using a policy dummy (*Treat* × *Post*) indicating pilot participation. *Treat* equals 1 if a firm's city is designated as a pilot city and 0 otherwise; *Post* equals 1 for 2018 and later, and 0 before. Table 11 reports that the interaction coefficients (*SCFD* × *DID*) are 0.0131 and 0.0237, both significant at the 5 % level. This result suggests that the impact of SCFD on corporate risk-taking is stronger in cities implementing the supply chain innovation pilot policy.

6. Conclusions

The global economic system is undergoing a profound reconfiguration, presenting enterprises and supply chains with an era of systemic transformation. Traditional SCF has long supported resource allocation and operational efficiency in supply chains. However, evolving market dynamics and technological advancements have exposed its limitations. Traditional SCF struggles to keep pace with the increasing complexity and dynamism of contemporary supply chain ecosystems and may inadvertently heighten systemic vulnerabilities. Considerably, the integration of digital technology has catalyzed a significant surge in demand for digital financial solutions, making SCF innovation a central focus. This transformation offers substantial opportunities for the advancement of SCFD systems.

This study examines the impact and mechanisms of SCFD on corporate risk-taking. Using a sample of China's, A-share listed firms from 2011 to 2023, the empirical findings demonstrate that SCFD enhances corporate risk-taking by easing financing constraints and reducing SCC. The effect is particularly significant in advancing the risk-taking of specific entities, including SMEs, growth-stage firms, and non-SOEs. Moreover, the positive association between SCFD and risk-taking is more pronounced under China's supply chain innovation pilot policy.

Although based on data from Chinese listed firms, these findings offer generalizable insights. Despite differences in political and economic systems, SCF practices share fundamental financial functions worldwide. Thus, the results provide meaningful policy implications for financial institutions, enterprises, and policymakers.

Table 10
Channel analysis.

Variable	Financial constraint			Supply chain concentration		
	(1)	(2)	(3)	(4)	(5)	(6)
	FC	*Risk1*	*Risk2*	*SCC*	*Risk1*	*Risk2*
SCFD	−0.0344**	0.0140***	0.0247***	−6.6543***	0.0149***	0.0262***
	(0.0152)	(0.0042)	(0.0074)	(1.3322)	(0.0044)	(0.0077)
FC		−0.0183***	−0.0302***			
		(0.0024)	(0.0041)			
SCC					0.0001***	0.0001***
					(0.0000)	(0.0001)
Constant	5.0021***	0.1943***	0.3118***	66.7057***	0.0975***	0.1542***
	(0.0626)	(0.0204)	(0.0358)	(6.2701)	(0.0171)	(0.0301)
Controls	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Observations	21,283	21,283	21,283	20,267	20,267	20,267
Adjusted R^2	0.879	0.382	0.384	0.763	0.379	0.381

Note: ***, **, and * demonstrate significance at the 1 %, 5 %, and 10 % levels, respectively.

Table 11
Moderating effect analysis.

Variable	(1) *Risk1*	(2) *Risk2*
SCFD	0.0115**	0.0197**
	(0.0045)	(0.0079)
*SCFD*DID*	0.0131**	0.0237**
	(0.0061)	(0.0106)
DID	−0.0003	−0.0003
	(0.0009)	(0.0016)
Constant	0.1063***	0.1647***
	(0.0177)	(0.0311)
Controls	Yes	Yes
Firm FE	Yes	Yes
Year FE	Yes	Yes
Observations	19,062	19,062
Adjusted R2	0.372	0.375

Note: ***, **, and * demonstrate significance at the 1 %, 5 %, and 10 % levels, respectively.

First, SCFD promotes risk-taking primarily by alleviating financing constraints. Advancing SCFD requires deep integration of frontier technologies, including big data, cloud computing, cross-chain protocols, and high-speed smart contracts. Developing innovative IoT–SCFD solutions and leveraging metaverse tools such as digital twinning and edge-cloud synergy will strengthen the digital infrastructure needed to overcome financing barriers for enterprises.

Second, SCFD encourages risk-taking by mitigating SCC. SCFD development should focus on decentralized, multilateral financial support networks that optimize nodal relationships and limit over-reliance on core enterprises. This includes establishing authenticated full-chain credit records, advancing data standardization and cross-chain interoperability and building multicentric platforms to eliminate information silos. Strengthening digital SCF infrastructure and enhancing the productive value of data are crucial for efficient allocation and cocreation among on-chain enterprises.

Third, the supply chain innovation pilot policy serves as a positive moderator in the SCFD–risk-taking relationship. Policymakers should gradually expand pilot scope and diversify SCFD programs to provide flexible financial support. Mobilizing large financial institutions as stabilizers and core firms as coimplementers of cross-regional SCFD platforms will increase participation. To strengthen incentives, tax benefits, regulatory support, and performance recognition should be offered to high-performing participants.

In conclusion, this study advances the literature by revealing how SCFD affects corporate risk-taking and the mechanisms involved. In China's context, the findings hold broad relevance for countries seeking to strengthen supply chain security and financial resilience. By clarifying how SCFD supports risk-taking behavior and economic outcomes, this research provides valuable guidance for international organizations and policymakers. These insights enable market participants to make informed decisions, enhance governance, and design innovative policies that bolster supply chain resilience in diverse economic environments.

Funding

This work was supported by the [The National Social Science Fund of China] under Grant [number 24XJY010].

Data availability

Data will be made available on request.

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